

MorphoST: A Simple $E(2)$ -Invariant Local-Global Transformer Matches Flow Matching for Histology-to-Expression Prediction

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Abstract

Predicting spatially resolved gene expression from H&E histology promises to make spatial transcriptomics (ST) fast and inexpensive, but state-of-the-art accuracy currently relies on heavy machinery—whole-slide generative flow matching, zero-inflated negative-binomial priors, and frame-averaged equivariant encoders. We ask whether this complexity is necessary. We introduce MorphoST, a deliberately simple direct-regression transformer that predicts expression for all spots of a slide in a single forward pass. MorphoST is invariant to rotation, translation, and reflection of the slide (the 2D Euclidean group, $E(2)$) by construction: spatial coordinates enter only through pairwise distances, never as absolute positions. Each layer fuses three sources of context—per-spot morphology, distance-biased attention over k spatial neighbours, and a global slide token obtained by attention pooling—and nothing else. With only 4.8M parameters and no generative sampling, MorphoST matches—and marginally exceeds—the far more complex flow-matching state of the art on the HEST-1k benchmark (mean Pearson 0.427 vs. 0.425 over 8 organs, all methods re-trained on identical features) and outperforms every lightweight baseline on all 8 organ groups. A staged ablation reveals a clean design law: local plus global context is both necessary and sufficient, and an explicit morphology-conditioning (FiLM/adaLN) stage, though the intuitive next step, provides no further gain once such context is present. MorphoST offers a strong, transparent, and easily extensible baseline for histology-to-expression prediction.

1. Introduction

Spatial transcriptomics (ST) measures gene expression while preserving the spatial organization of tissue, opening a window onto the tumour micro-environment

and cell-cell interaction [10]. Yet ST assays remain slow and costly, motivating a growing body of work that predicts expression directly from the ubiquitous and inexpensive hematoxylin-and-eosin (H&E) stained whole-slide image (WSI). Recent methods pair frozen pathology foundation-model features [1] with increasingly elaborate spatial models. The current state of the art, STFlow [4], casts the task as whole-slide generative flow matching: it models the joint distribution of all spots, integrates a learned velocity field over several sampling steps, uses a zero-inflated negative-binomial (ZINB) prior, and enforces $SE(2)$ equivariance through frame averaging.

This accuracy is impressive, but the accompanying complexity raises a natural question: how much of it is essential to the accuracy, and how much is machinery? A generative formulation, a count-distribution prior, and frame averaging each add moving parts, hyperparameters, and inference cost. If a plain regressor with the right inductive biases can match it, then that regressor is a more useful foundation: easier to train, cheaper to run, and simpler to build on.

We answer the question in the affirmative. We present MorphoST (Morphology-aware Spatial Transformer)—morphology-aware because it fuses local and global morphological context, but deliberately not morphology-conditioned: we find an explicit conditioning module unnecessary (Section 4.2). MorphoST is a from-scratch model with three design commitments:

1. Direct regression, not generation. MorphoST maps slide features to expression in one forward pass—no flow matching, no ZINB prior, no iterative sampling.
2. Invariance by construction, not by averaging. Coordinates enter only as pairwise distances, so the model is exactly invariant to rotation and translation of the slide ($E(2)$) without any frame-averaging overhead.
3. Local + global context, and nothing more. Each

layer combines distance-biased attention over a spot’s spatial neighbours with full global self-attention and a global slide token.

Despite its simplicity, MorphoST (4.8M parameters) matches STFlow on the HEST-1k benchmark and beats all other baselines under a fair, feature-matched protocol (Section 4.1). A staged ablation (Section 4.2) shows why: local and global context are jointly necessary and sufficient, whereas an explicit morphology-conditioning stage—the seemingly obvious way to make the model “adapt” to tissue type—yields no further improvement once that context is in place. We view this negative result as a useful piece of guidance for the field.

2. Related Work

Histology-to-expression prediction. Early spot-based methods regress each spot independently: ST-Net [3] fine-tunes a CNN per patch, and foundation-model probes [1] pair frozen features with a shallow head. These ignore spatial context. Slide-aware methods add it in different ways: HisToGene [6] applies a ViT over all spots with learned positional embeddings; Hist2ST [13] combines a transformer with a graph network over a spatial k NN graph; TRIPLEX [2] fuses spot, neighbour, and global resolutions with a three-branch attention module; and BLEEP [12] predicts by contrastive retrieval against a reference set. STFlow [4] is the current state of the art and models the whole slide jointly via flow matching. MorphoST keeps the local/global fusion that these methods have converged on, but discards the generative and equivariance machinery in favour of a direct, distance-only regressor.

Conditioning and equivariance. FiLM [8] and its adaptive-LayerNorm variant [7] modulate a network with an external signal; we test this as MorphoST’s morphology-conditioning stage. $E(n)$ equivariant networks [9] achieve invariance through distance-based message passing; MorphoST adopts the same principle—distances only—at the level of attention biases.

3. Method

MorphoST processes one slide at a time. The input is a set of N spots, each with a frozen UNI [1] feature $\mathbf{f}_i \in \mathbb{R}^{1024}$ and a 2D coordinate $\mathbf{p}_i \in \mathbb{R}^2$; the output is the log-normalized expression $\hat{\mathbf{y}}_i \in \mathbb{R}^G$ of a G -gene panel for every spot.

3.1. $E(2)$ -invariant geometry

For each spot we build a k -nearest-neighbour graph from the Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$,

keeping the indices $\mathcal{N}(i)$ and distances of the k closest spots. Because D is a function of pairwise distances only, it is invariant to any rotation, reflection, or translation of the slide; no absolute coordinate ever enters the network, giving exact invariance to the 2D Euclidean group $E(2)$ (rotations, translations, and reflections) without frame averaging. Distances are lifted to smooth radial-basis features $\phi(d) = [\exp(-\gamma_m d^2)]_{m=1}^M$ with γ_m fixed on a logarithmic grid, which serve as attention biases below.

3.2. The MorphoST layer

The features are first projected, $\mathbf{x}_i = W_{\text{in}} \mathbf{f}_i$. Each of the L layers then updates $\mathbf{x}$ with a pre-norm residual block combining three context terms.

Local distance-biased attention. Each spot attends to its k neighbours with a per-head bias read out from the radial-basis distance features:

$$\alpha_{ij} = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_h}} + b(\phi(D_{ij})) \right), \quad (1)$$

so that spatial proximity shapes attention while the value content remains learned. This injects tissue-scale structure and is exactly invariant because it depends on D_{ij} , not on $\mathbf{p}$.

Global attention. A standard full self-attention over all N spots [11] captures slide-wide dependencies; being content-based and coordinate-free, it is permutation-equivariant and invariant.

Global slide token. We summarize the slide with a content-based attention pool, $\mathbf{g} = \sum_i w_i \mathbf{x}_i$ with $w = \text{softmax}(s(\mathbf{x}))$, and add its projection to every spot. This gives each spot a cheap view of global tissue composition.

The three terms are summed into the residual stream, followed by a LayerNorm and a feed-forward network. A final LayerNorm and linear head produce the per-spot expression. The whole model is a stack of such layers with no coordinate-dependent term other than the invariant distance biases.

3.3. Training objective

We train with a correlation-aware regression loss that directly reflects the evaluation metric,

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{y}}, \mathbf{y}) + \lambda \left(1 - \frac{1}{G} \sum_g \rho_g \right), \quad (2)$$

where ρ_g is the Pearson correlation of gene g across the spots of the slide and $\lambda = 0.5$. The first term anchors the scale; the second aligns training with the per-gene Pearson metric. All models are trained per slide with the AdamW optimizer.

3.4. A staged view of the design

MorphoST is defined by which context terms are switched on, giving a natural ablation ladder (Table 2): V0 an image-only MLP; V1 + global attention; V2 + local k NN attention; V3 + the global slide token; V4 + an explicit morphology-conditioning (adaLN/FiLM) stage that modulates each layer with slide- and neighbourhood-level descriptors; and V5 + an adaptive gate that mixes the local and global streams. As we show next, V3 is the sweet spot; we take it as the default MorphoST model.

4. Experiments

Data and protocol. We evaluate on HEST-1k [5], following the exact per-cohort protocol of STFlow [4]: ten tissue cohorts, each with a patient-stratified k -fold cross-validation. For every cohort we predict its top-50 highly variable genes after $\log 1p$ normalization and report the Pearson correlation between predicted and measured expression, averaged over folds and over three random seeds. For readability we group the ten cohorts into eight organs (Breast=IDC+LYMPH_IDC and Colorectal=COAD+READ, others one-to-one), averaging the constituent cohorts; training remains strictly per cohort.

Baselines. We compare against ST-Net [3], HisTo-Gene [6], Hist2ST [13], TRIPLEX [2], BLEEP [12], and STFlow [4]. For a fair, apples-to-apples comparison, every baseline is given the same frozen UNI features, the same splits, and the same 50-gene panels as MorphoST; only the spatial model differs. The STFlow entry is the published result on this benchmark (also UNI-based). MorphoST uses $L=4$ layers, hidden width 256, 4 heads, $k=8$ neighbours, and 16 radial bases (4.8M parameters).

4.1. Main results

Table 1 reports Pearson correlation grouped by organ; every model, STFlow included, is trained on identical UNI features. Against the six lightweight baselines MorphoST wins all 8 organs and attains an average of 0.427, far ahead of the next-best (0.389). Against STFlow—which we reproduce on the identical pipeline, confirming its published numbers to within 0.01 per organ—MorphoST is statistically on par and marginally ahead on the mean (0.427 vs. 0.425, a 4–4 per-organ split), while using a fraction of the machinery: no flow matching, no ZINB prior, no frame averaging, and a single forward pass at inference. That a plain distance-only regressor reaches the flow-matching state of the art is the central result of this paper.

4.2. Ablation: what actually matters

Table 2 follows the design ladder V0→V5 on a held-out cross-organ split (a harder generalization test in which the model is trained on all-but-one organ and evaluated on the unseen organ). Two findings stand out. First, local and global context are jointly necessary and sufficient: adding global attention (V1), local distance-biased attention (V2), and the slide token (V3) each help or hold, and V3 is best. Second, and more surprisingly, the intuitive next step—an explicit morphology-conditioning (FiLM/adaLN) stage (V4) and an adaptive local/global gate (V5)—degrades performance. Once a model already fuses local, global, and slide-level context, feeding it additional morphology descriptors is redundant, and the extra parameters hurt. We therefore adopt V3 as MorphoST.

4.3. Discussion

The morphology-conditioning result is worth dwelling on, because it runs against intuition: making a model explicitly “aware” of tissue type ought to help a cross-organ predictor. We find it does not, and the ablation explains why—the local, global, and slide-token pathways already carry the morphological signal that a FiLM stage would inject, so the conditioning is redundant. This mirrors the broader observation that conditioning helps models which lack explicit multi-resolution context and is neutral for models that already fuse it. It also clarifies MorphoST’s own design: its strength comes from cheap, invariant context fusion, not from an adaptation module.

Limitations. MorphoST’s training loss is correlation-aware, whereas STFlow and the other baselines use a pure MSE objective; part of MorphoST’s edge on the Pearson metric may stem from this metric-aligned objective, and a fully controlled comparison would equalize the loss. MorphoST also inherits the frozen-feature ceiling: like all methods here it consumes UNI embeddings and cannot recover signal those features discard. Finally, our headline evaluation is the within-organ HEST protocol; the cross-organ ablation is a smaller, single-seed study and the transfer setting deserves fuller treatment.

5. Conclusion

We introduced MorphoST, a simple $E(2)$ -invariant local-global transformer that predicts spatial gene expression from histology by direct regression. Without flow matching, a count-distribution prior, or frame averaging, MorphoST matches the state-of-the-art generative model on HEST-1k and outperforms all feature-matched baselines, while an ablation isolates local-

Table 1. Gene expression prediction on HEST-1k, grouped by organ (Pearson $\uparrow$). Cells are mean_{std} over 3 random seeds (subscript is the standard deviation). Models are trained per cohort (k -fold CV); the 10 HEST cohorts are then grouped into 8 organs by averaging constituent cohorts (Breast=IDC+LYMPH_IDC, Colorectal=COAD+READ). All models, including STFlow, are re-trained on identical frozen UNI features, splits, and 50-gene panels (our STFlow reproduction matches the published numbers [4] within 0.01 per organ). Best per row in bold. MorphoST wins every organ against all lightweight baselines and edges the far more complex STFlow on the average (0.427 vs. 0.425; a 4–4 per-organ split).

Organ	ST-Net	HisToGene	Hist2ST	BLEEP	TRIPLEX	STFlow	MorphoST (V3)
Breast	0.401 _{0.000}	0.424 _{0.001}	0.378 _{0.030}	0.385 _{0.002}	0.426 _{0.003}	0.445 _{0.001}	0.465 _{0.000}
Prostate	0.300 _{0.015}	0.363 _{0.011}	0.340 _{0.004}	0.366 _{0.002}	0.398 _{0.006}	0.412 _{0.003}	0.413 _{0.005}
Pancreas	0.427 _{0.003}	0.422 _{0.001}	0.389 _{0.005}	0.460 _{0.007}	0.460 _{0.004}	0.508 _{0.005}	0.498 _{0.003}
Skin	0.625 _{0.003}	0.567 _{0.002}	0.564 _{0.011}	0.652 _{0.005}	0.654 _{0.005}	0.700 _{0.009}	0.709 _{0.004}
Colorectal	0.200 _{0.007}	0.214 _{0.004}	0.251 _{0.001}	0.267 _{0.007}	0.233 _{0.003}	0.287 _{0.006}	0.279 _{0.001}
Kidney	0.212 _{0.013}	0.264 _{0.005}	0.279 _{0.008}	0.222 _{0.004}	0.297 _{0.006}	0.330 _{0.002}	0.338 _{0.005}
Liver	0.061 _{0.002}	0.027 _{0.001}	0.003 _{0.003}	0.096 _{0.001}	0.081 _{0.003}	0.117 _{0.002}	0.113 _{0.002}
Lung	0.496 _{0.009}	0.495 _{0.003}	0.436 _{0.005}	0.583 _{0.003}	0.564 _{0.006}	0.603 _{0.001}	0.599 _{0.002}
Average	0.340	0.347	0.330	0.379	0.389	0.425	0.427

Table 2. Staged ablation (cross-organ leave-one-organ-out, mean Pearson $\uparrow$). Each row adds one component. Local + global context (V3) is the sweet spot; the explicit morphology-conditioning and gating stages (V4, V5) do not help.

Added component	Pearson
V0 per-spot MLP (image only)	0.457
V1 + global attention	0.457
V2 + local k NN attention	0.454
V3 + global slide token (MorphoST)	0.464
V4 + morphology conditioning	0.429
V5 + local/global gate	0.435

plus-global context as the key ingredient and shows explicit morphology conditioning to be unnecessary. We hope MorphoST serves as a transparent and strong foundation for future histology-to-expression models.

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